

AI-PAL: Self-Supervised AI Phase Picking via Rule-based Algorithm for Generalized Earthquake Detection

Yijian Zhou^{1,3*}, Hongyang Ding², Abhijit Ghosh¹, and Zengxi Ge²

¹Department of Earth and Planetary Sciences, University of California, Riverside, USA

²Institute of Theoretical and Applied Geophysics, Peking University, Beijing, China

³Division of Geological and Planetary Sciences, California Institute of Technology, Pasadena, CA, USA

Contents of this file

Text S1 to S2

Figures S1 to S25

Tables S1 to S2

Dataset S1 to S4 (uploaded separately)

Introduction

This supporting material provides additional information on the data condition, details in data cleaning algorithms, and data visualizations for the catalogs mentioned in the main text.

Text S1.

Details on data cleaning strategies. For both the PAL and SAR picker, we add a rule-based algorithm to help fill short data gaps and to remove pulse-like data glitches. For PAL, the glitch removal algorithm starts from v3.2. For the data gap removal, we define a data gap as the data with the same amplitude values (typically 0) for a consecutive points. For the gap within a length of 5 s, we substitute the gap period with the data of the same length afterwards (Figure S3). This gap filling process is especially important for SAR, because after filtering, the data gaps will produce pulse-like noises that fools AI pickers. For longer data gaps, we do not apply special treatments in PAL, but will discard the sliding window if the gap period exceeds half of the window length in SAR.

For data glitch removal, we design criteria based on amplitude-ratio (Figure S4). We define three amplitude ratios (Figure S4a): (1) $R1=a1/a2$, where $a1$ is the amplitude of the first wiggle after P or S arrival, and $a2$ is that of the second wiggle; (2) $R2=A1/A2$, where $A1$ is the P-wave amplitude, measured between t_p to $t_p+(t_s-t_p)/2$, and $A2$ is the P-tail amplitude, measure between $t_p+(t_s-t_p)/2$ to t_s ; and (3) $R3=A1/A3$, where $A3$ is the S-wave amplitude, measured between t_s to $t_s+(t_s-t_p)/2$. Note that we use the minimum value of the 3-channel for all amplitude ratios, and further select the maximum value of P and S wave for $R1$. We specifically deal with three types of glitches that show different patterns of amplitude ratio (Figure S4b-d). By comparing the distribution of these amplitude ratios before and after phase association (Figure S5), we can see a clear separation between glitches and earthquakes in all the amplitude ratios, and that the Ridgecrest-Coso data exhibit more data glitches compared to the EAFZ case. This distribution also helps us set a proper default value as $R1<5$, $R2<8$, and $R3<3$. By applying the above data glitch removal algorithm, we can see a ~30% decrease in false picks in the Ridgecrest-Coso case (Figure S6).

Text S2.

Updates on PAL association algorithm. Starting from v3.0, PAL made an update in the phase association algorithm, aiming at large-scale applications or extremely intense sequences. In the association process, PAL first group clusters of origin time estimations, and process each cluster by examine the P wave travel time residual. For large-scale applications, or when the seismicity rate is extremely high, one original time cluster make contain events from different locations. However, in the previous algorithm (Zhou et al., 2021), each cluster can only produce one event detection, because the picks with a too large P residual will be discarded. In the new algorithm, we allow the discarded picks to join into the cluster again, and each pick can be recalled for a maximum number of times. For the two cases in this study, this parameter is set as 1. Note that we still do not allow a pick to appear in two different events. In this way, the new PAL associator is more scalable.

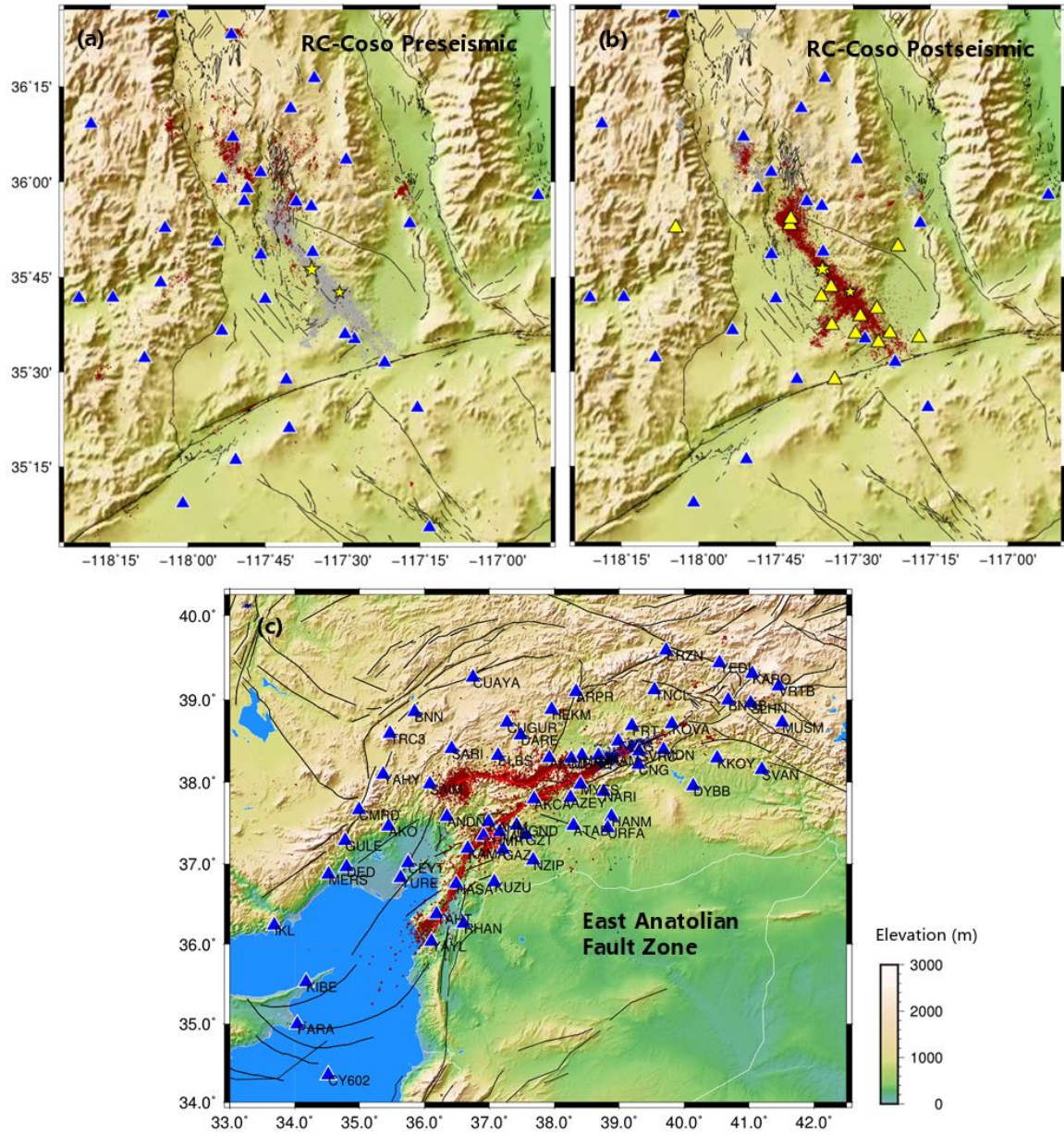


Figure S1. Distribution of seismic stations. (a-b) plot Ridgecrest-Coso (RC) region preseismic period (2008-2019/06) and aftershock period (2019/07/04-24). The blue triangles are broad-band and short-period seismometers, and yellow triangles are temporary stations deployed after the 2019 Ridgecrest earthquake (Hauksson et al., 2020). (c) plot the East Anatolian Fault Zone (EAFZ) pre-postseismic period (2020-2023/04).

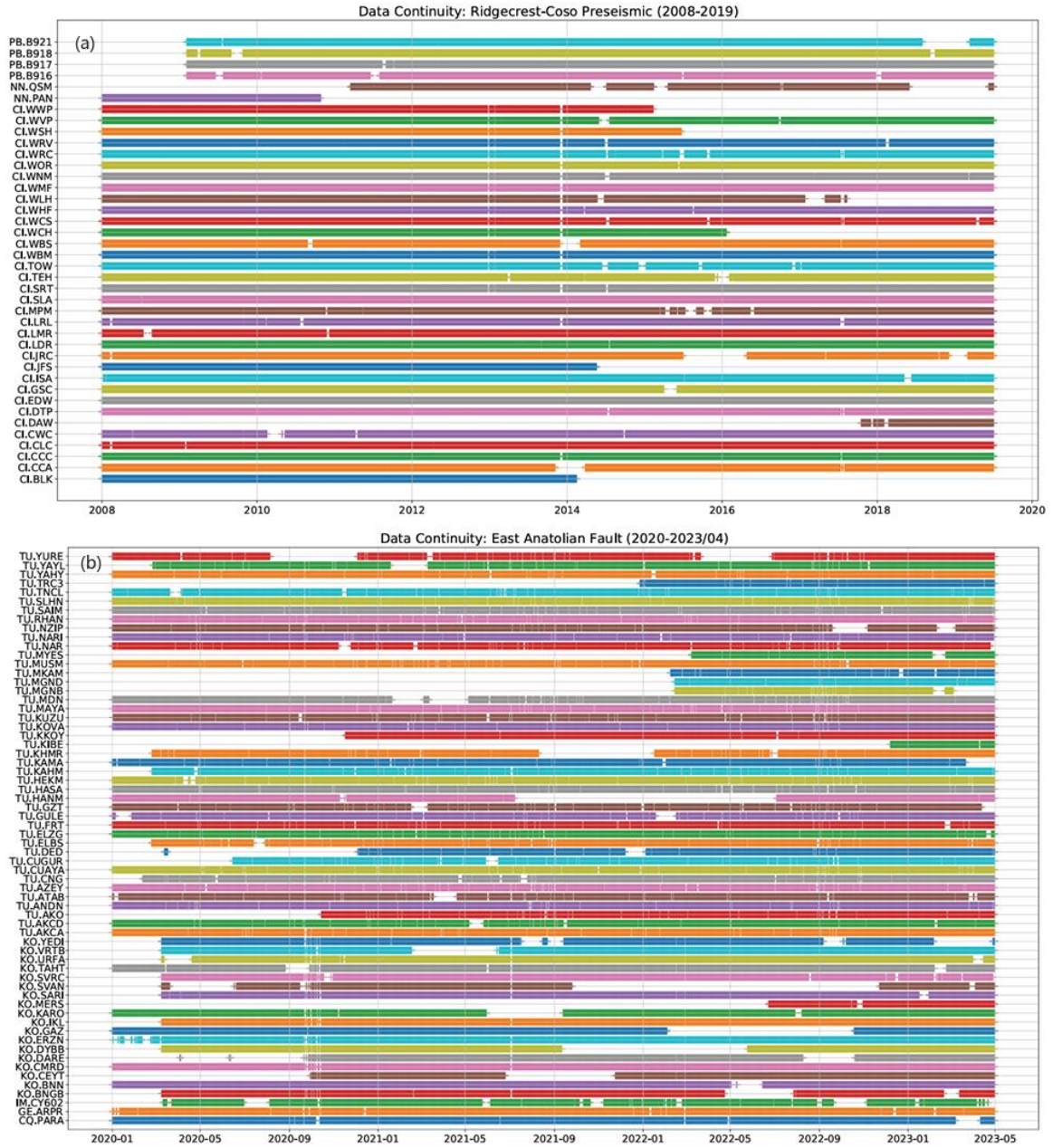


Figure S2. Data continuity. (a-b) plot Ridgecrest-Coso region preseismic period (2008-2019/06) and the East Anatolian Fault Zone pre-postseismic period (2020-2023/04), respectively. The station-date pairs with data available are marked by "+".

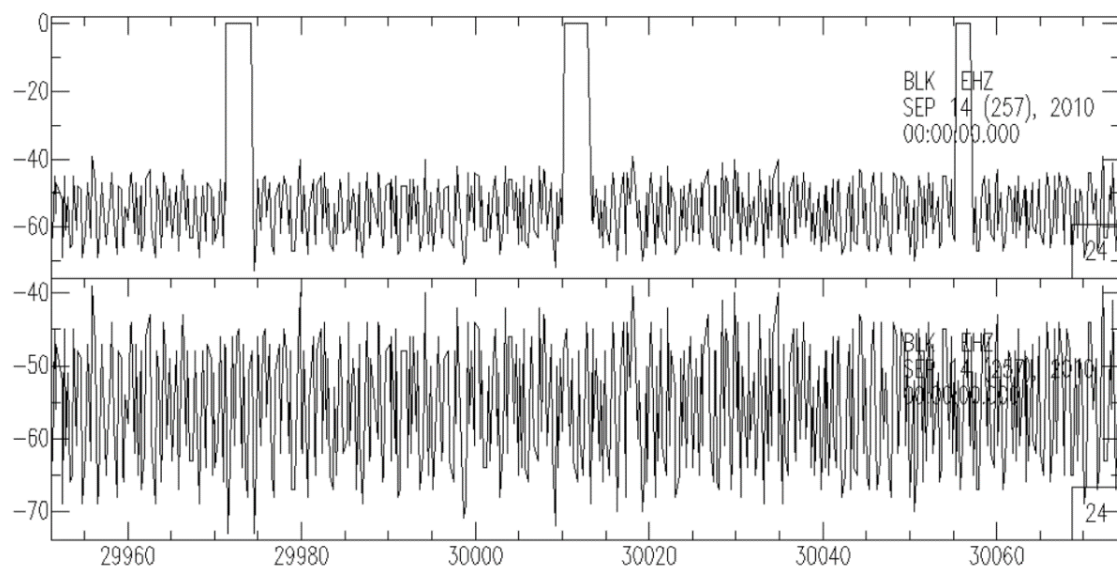


Figure S3. Example data gap and filling method. The upper and bottom panel show data before and after data gap filling.

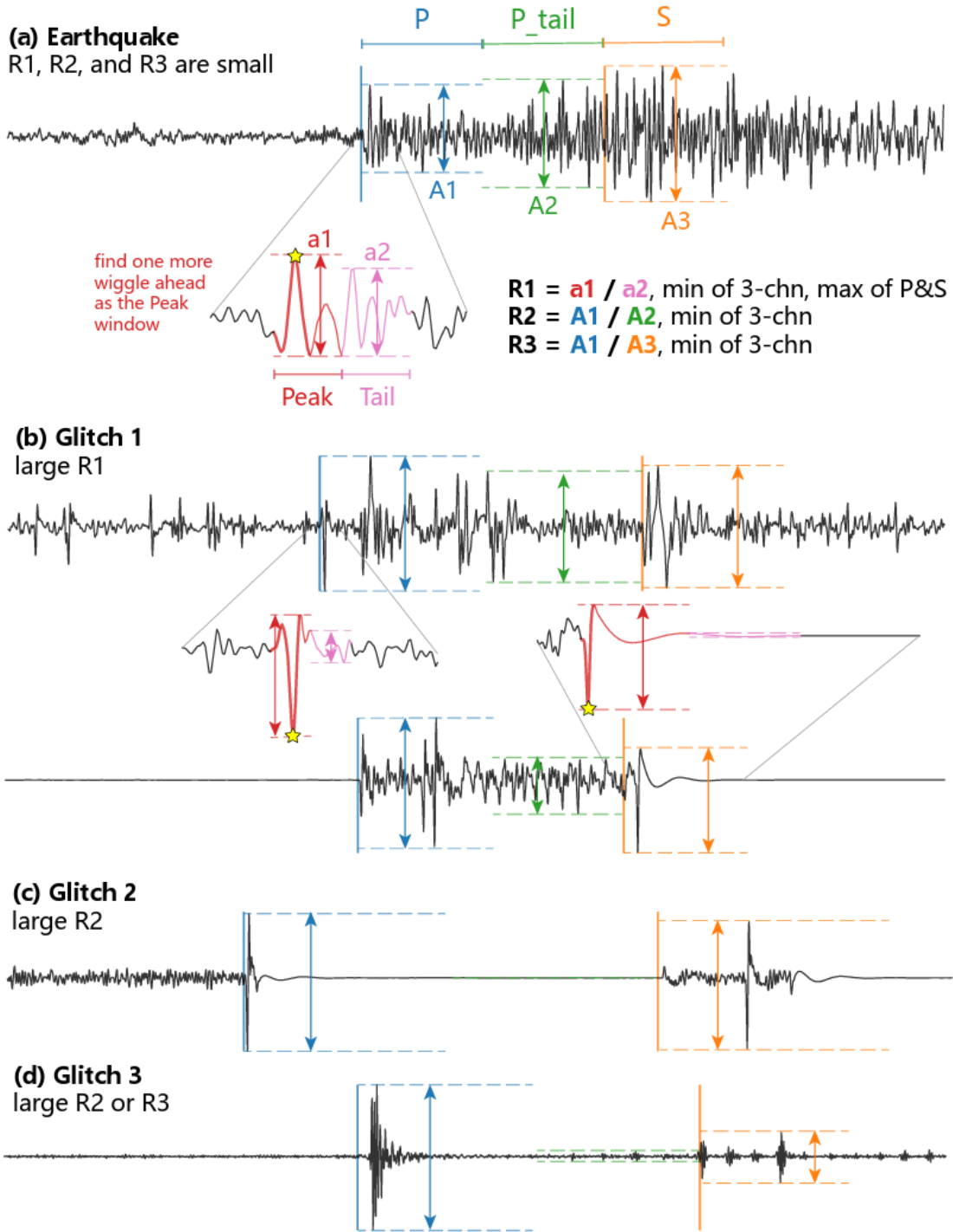


Figure S4. Definition of amplitude ratios used in the glitch removal algorithm. (a) show the typical amplitude ratio for earthquakes; (b-d) show amplitude ratio for three types of data glitches.

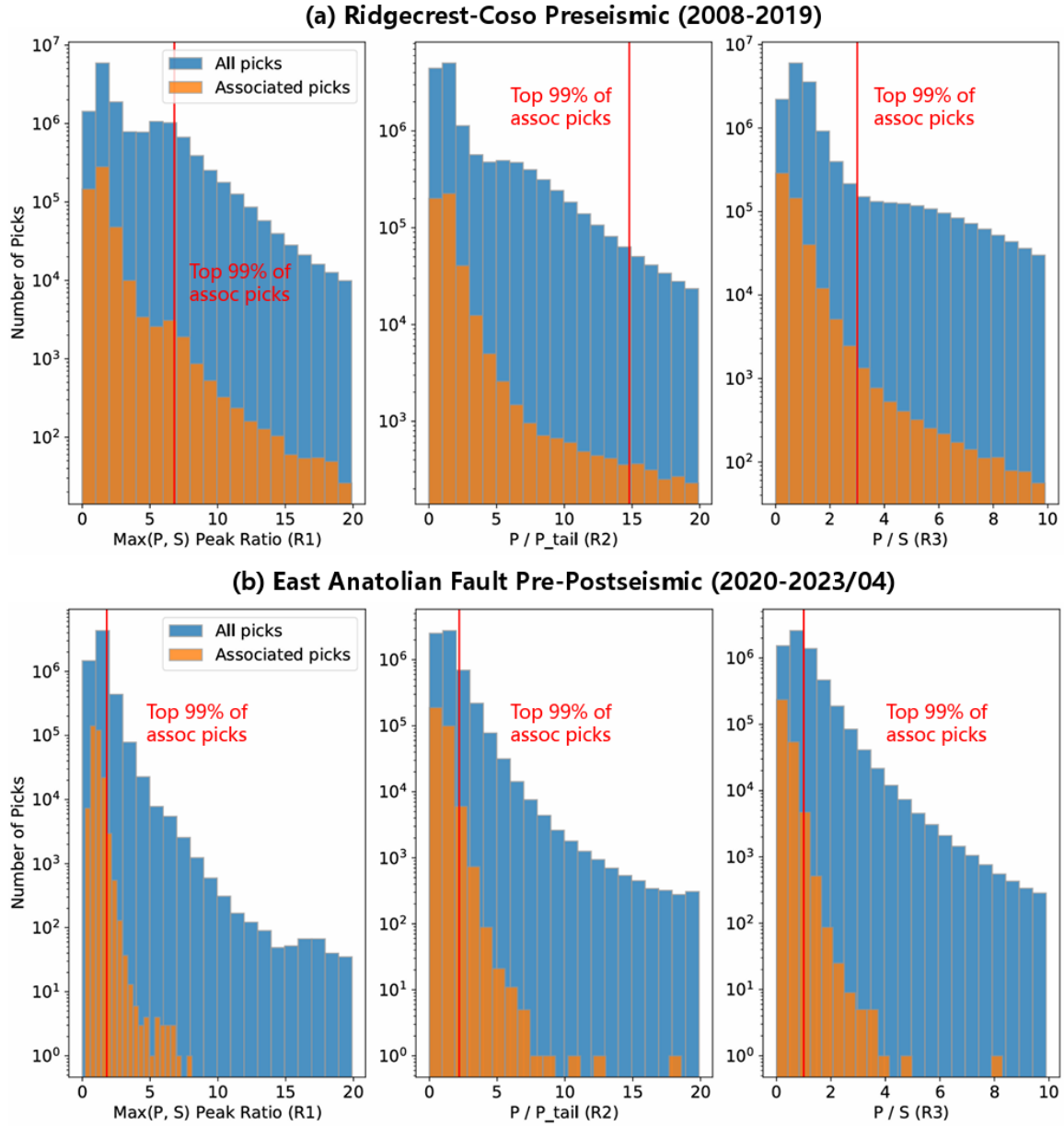


Figure S5. Distribution of amplitude ratios in (a) Ridgecrest-Coso preseismic period (2008-2019/06) and (b) East Anatolian Fault Zone pre-post-seismic period (2020-2023/04). The blue and orange bars denote all picks and associated picks, respectively. The vertical lines show the value for top 99% amplitude ratio in associated picks.

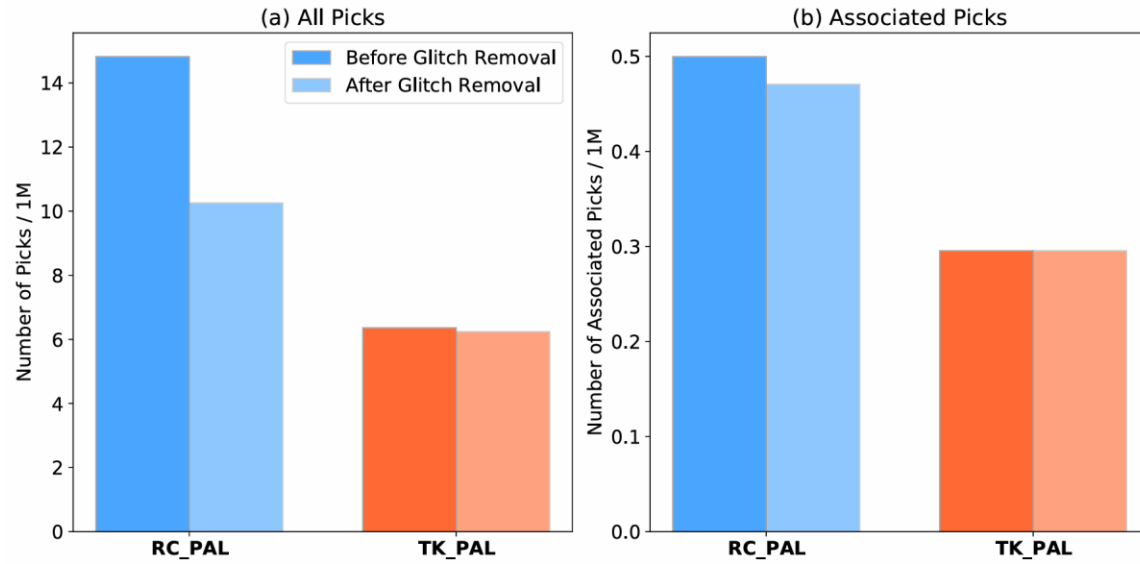


Figure S6. Number of all picks and associated picks before and after applying the amplitude ratio-based glitch removal algorithm. (a) plot number of all picked P&S pairs generated in Ridgecrest-Coso (RC) preseismic period (2008-2019/06) and EAFZ, Turkey (TK) pre-postseismic period (2020-2023/04). (b) plot number of associated P&S picks. Darker colors denote the results before applying the amplitude-ratio-based glitch removal algorithm, and lighter colors denote that after applying glitch removal algorithm.

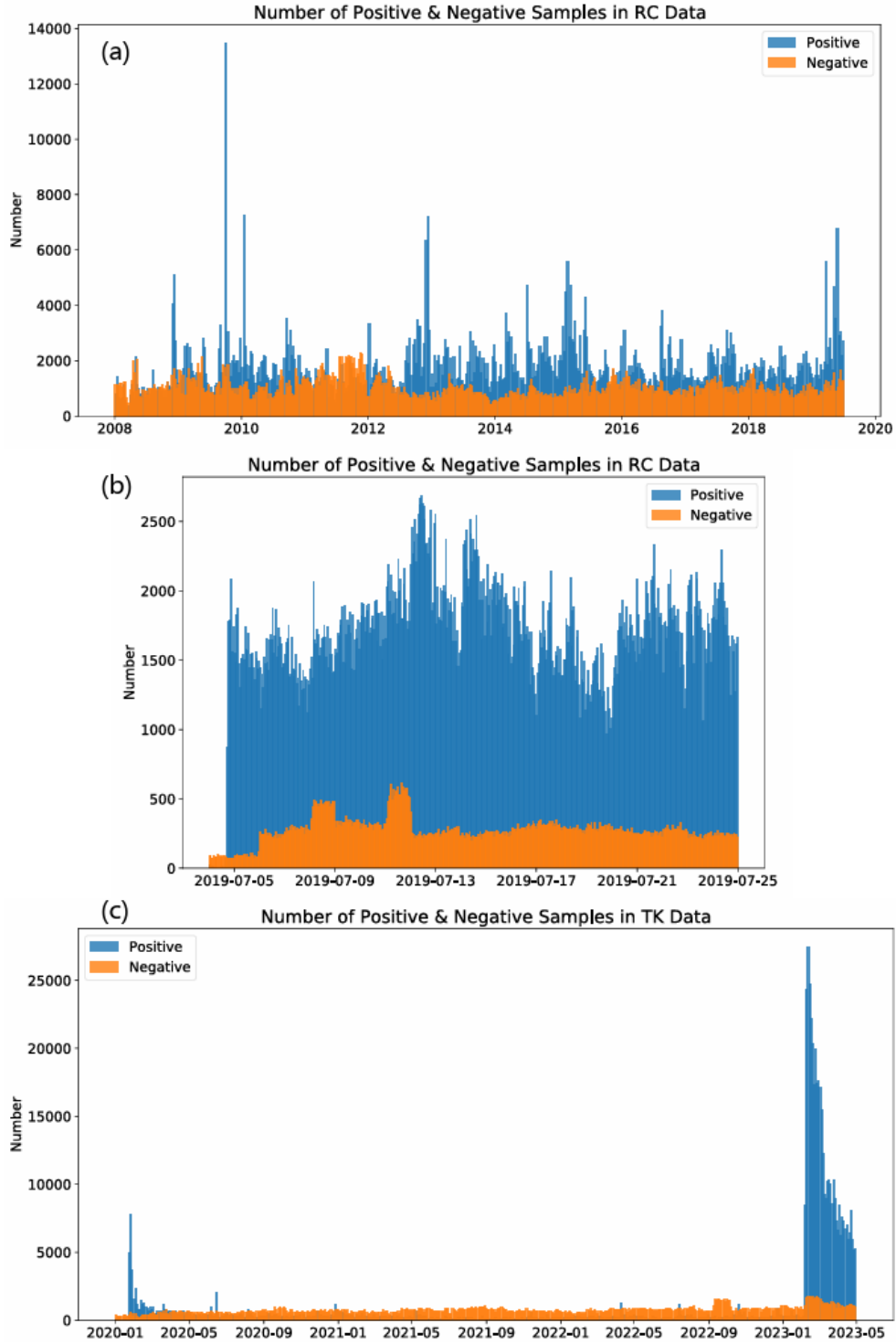


Figure S7. Number of training samples for different periods of time. Blue and orange bars denote the number of positive and negative samples, respectively. (a) for Ridgecrest-Coso preseismic period (2008-2019/06), (b) for Ridgecrest-Coso aftershock period (2019/07/04-24), and (c) for EAFZ pre-postseismic period (2020-2023/04).

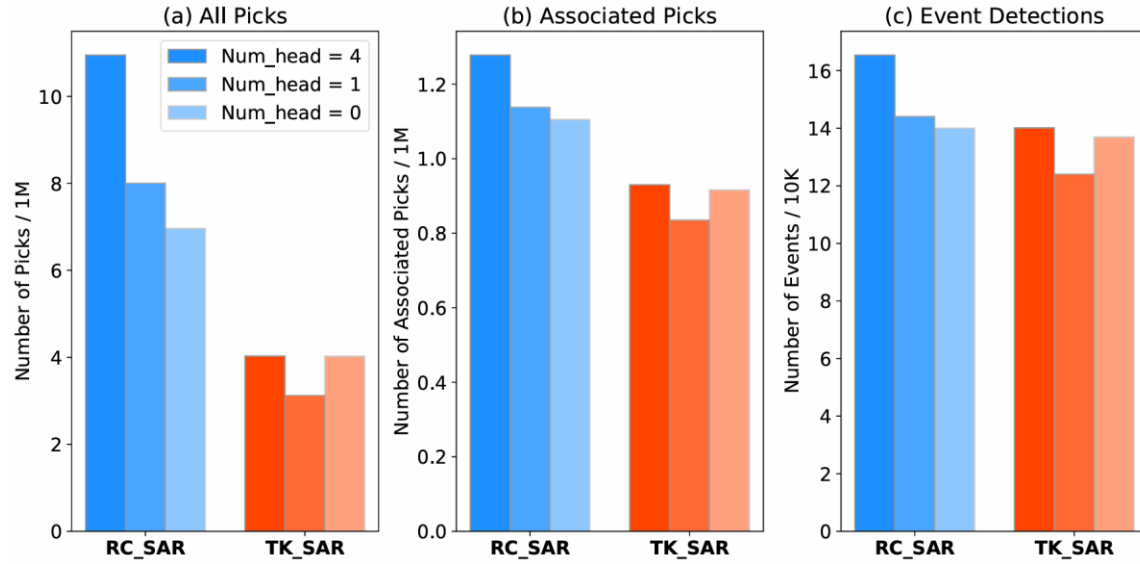


Figure S8. Comparison of phase detection performance with different numbers of self-attention heads (Num_head). (a) plot number of all picked P&S pairs generated in Ridgecrest-Coso (RC) preseismic period (2008-2019/06) and EAFZ, Turkey (TK) pre-postseismic period (2020-2023/04). (b) plot number of associated P&S picks. (c) plot the number of associated events.

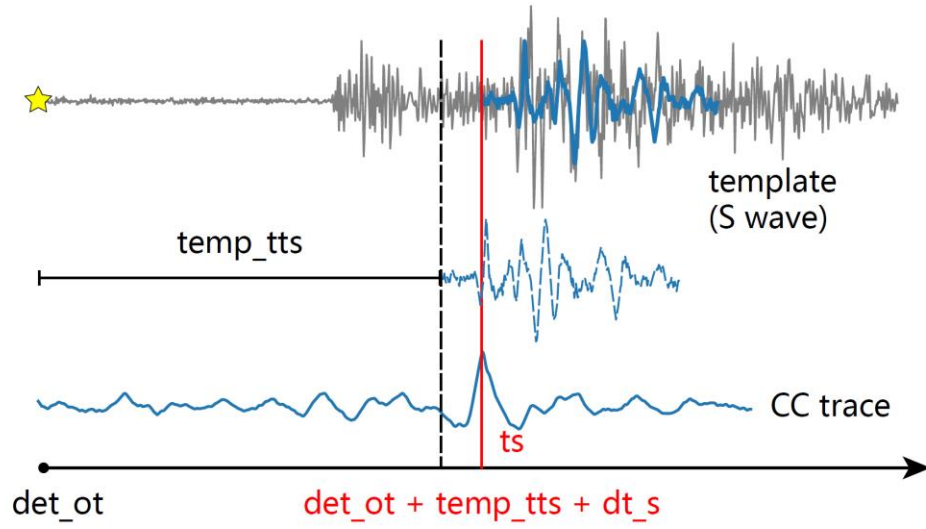


Figure S9. Demonstration of differential travel time measurement with waveform cross-correlation. The two events are named as "detection" and "template", travel time and differential travel time are abbreviated as "tt" and "dt", respectively.

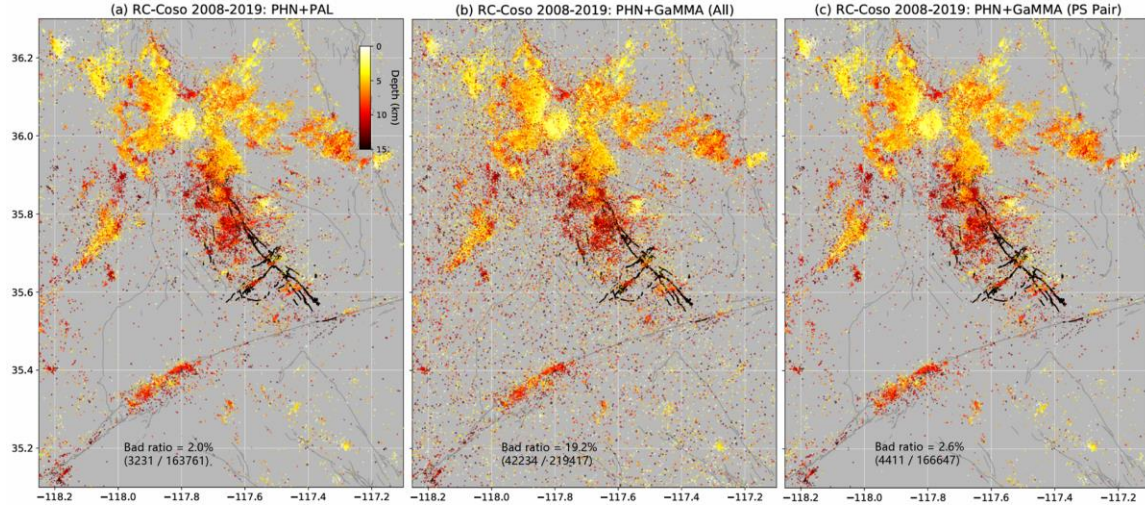


Figure S10. Comparison of location results with different association algorithms for RC-Coso preseismic period (2008-2019/06). (a-c) plot the map-view distribution of the hypolInverse location results, where the input phase files are generated by PhaseNet (PHN) + PAL associator, PHN + GaMMA (all picks), and PHN + GaMMA (only paired P&S picks), respectively.

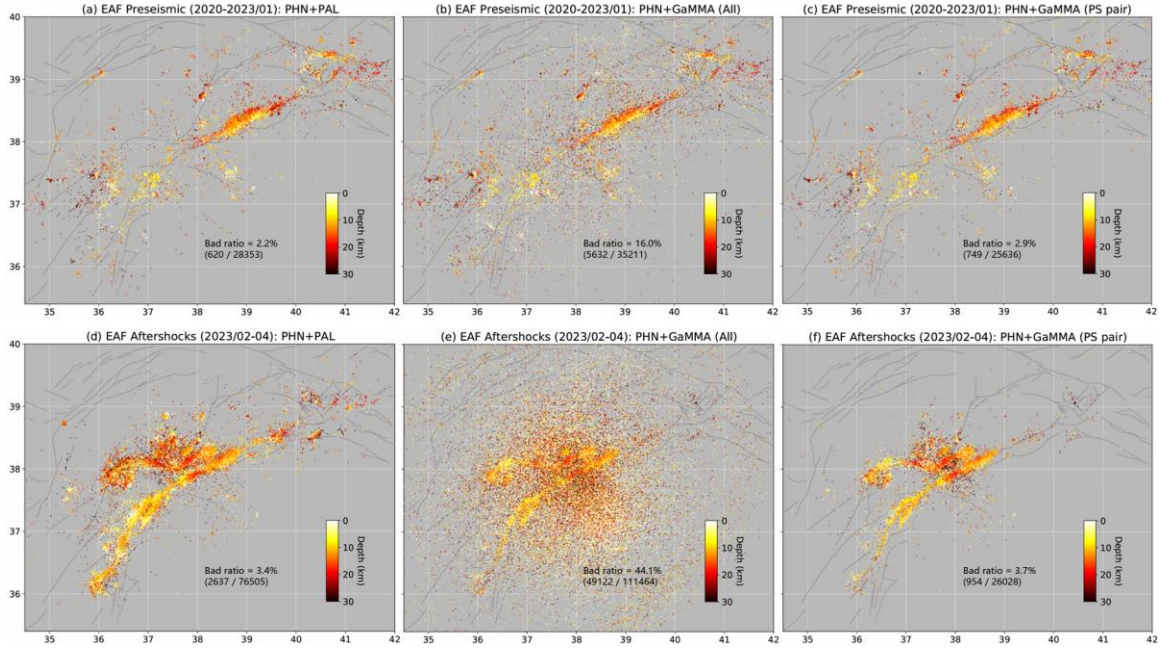


Figure S11. Comparison of location results with different association algorithms for EAF pre-postseismic period (2020-2023/01). (a-f) plot the map-view distribution of the hypolInverse location results, where the input phase files are generated by PhaseNet (PHN) + PAL associator, PHN + GaMMA (all picks), and PHN + GaMMA (only paired P&S picks), respectively. (a-c) plot the preseismic period (2020-2023/01) and (d-f) plot the aftershock period (2023/02-04).

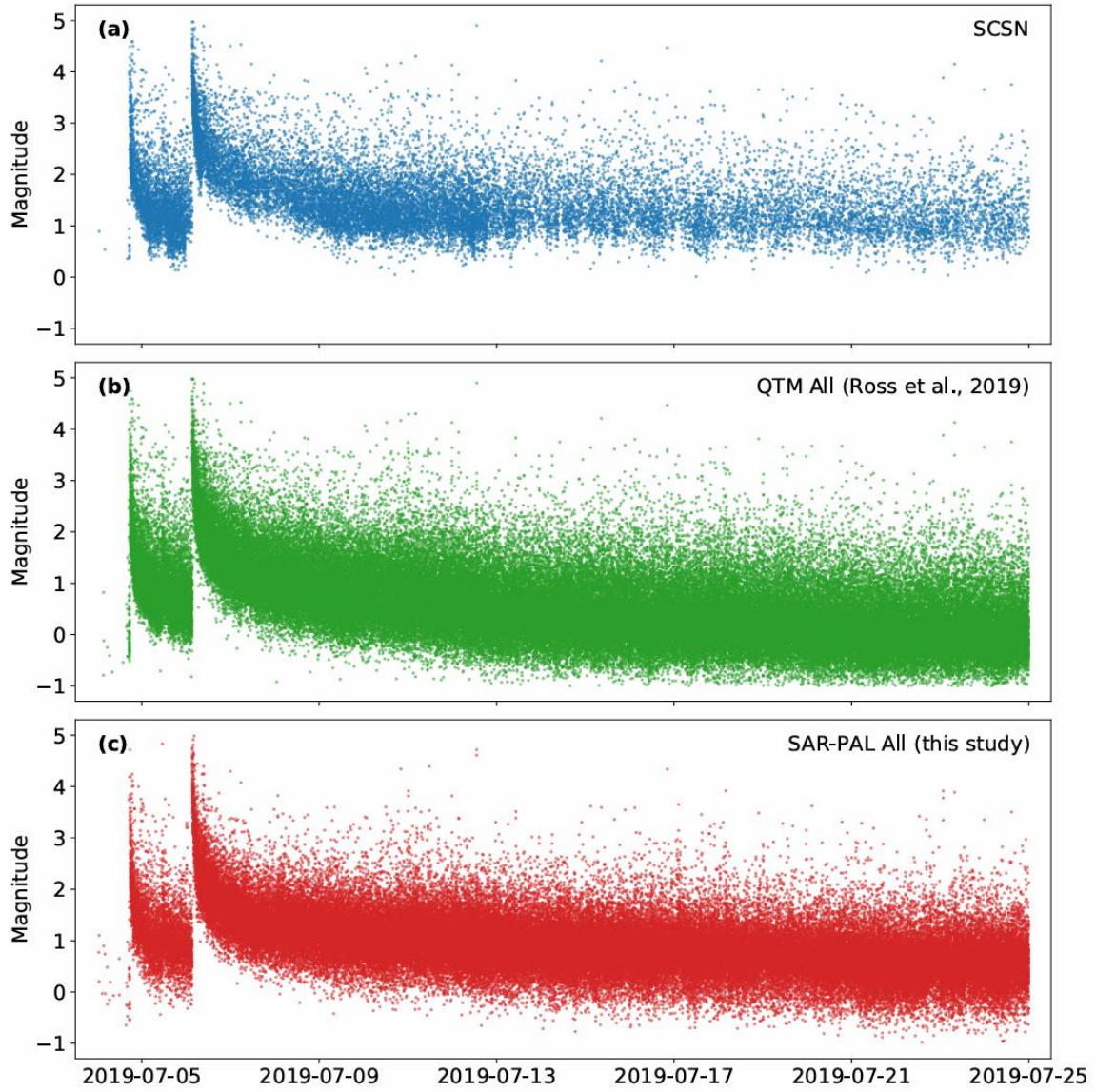


Figure S12. Magnitude-time sequence for Ridgecrest-Coso aftershocks shown in Figure 5a. (a-c) plot the SCSN catalog, QTM catalog, and SAR-PAL catalog, respectively.

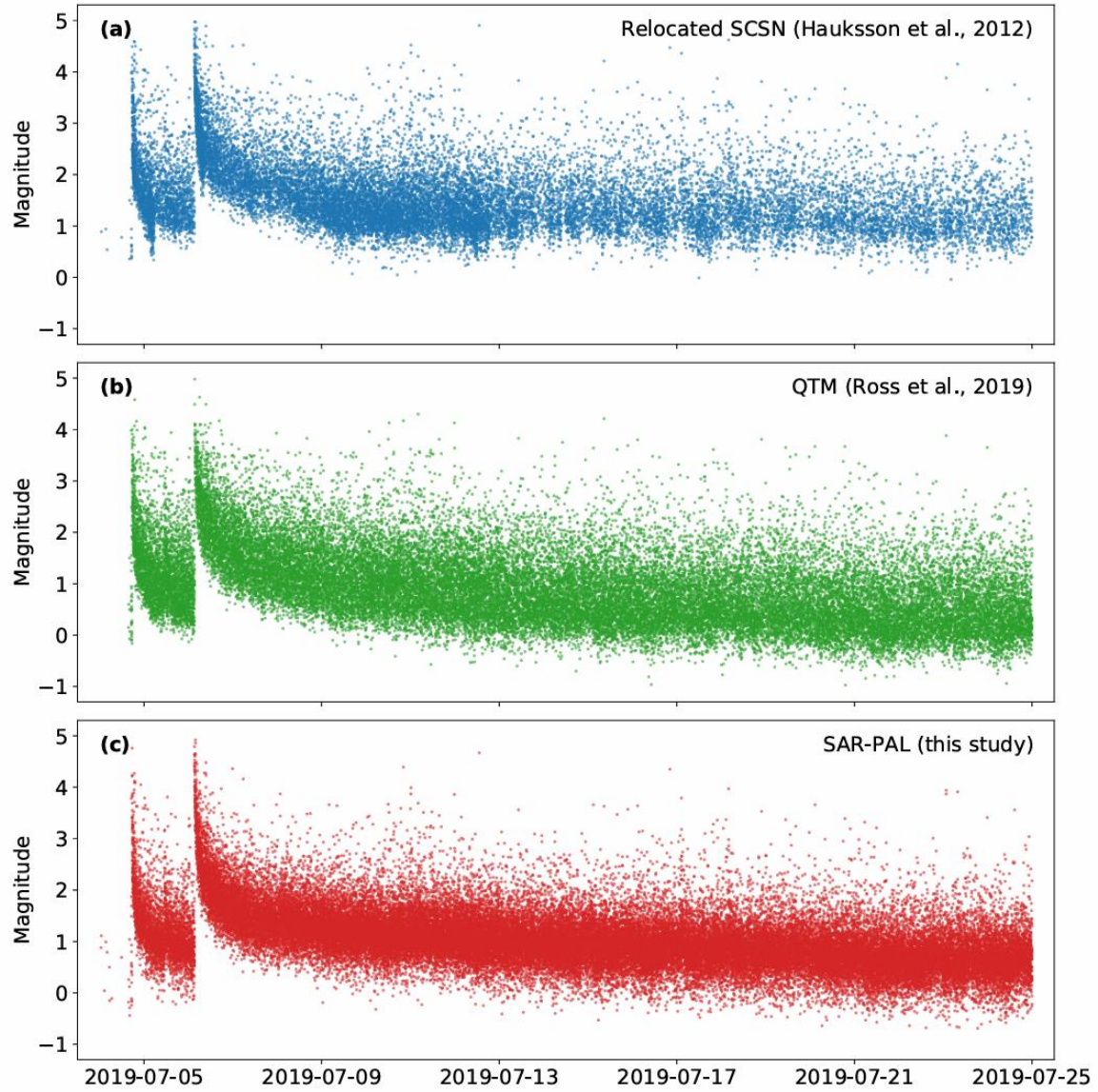


Figure S13. Magnitude-time sequence for Ridgecrest-Coso aftershocks shown in Figure 5b. (a-c) plot the relocated SCSN catalog, well-located QTM catalog, and relocated SAR-PAL catalog, respectively.

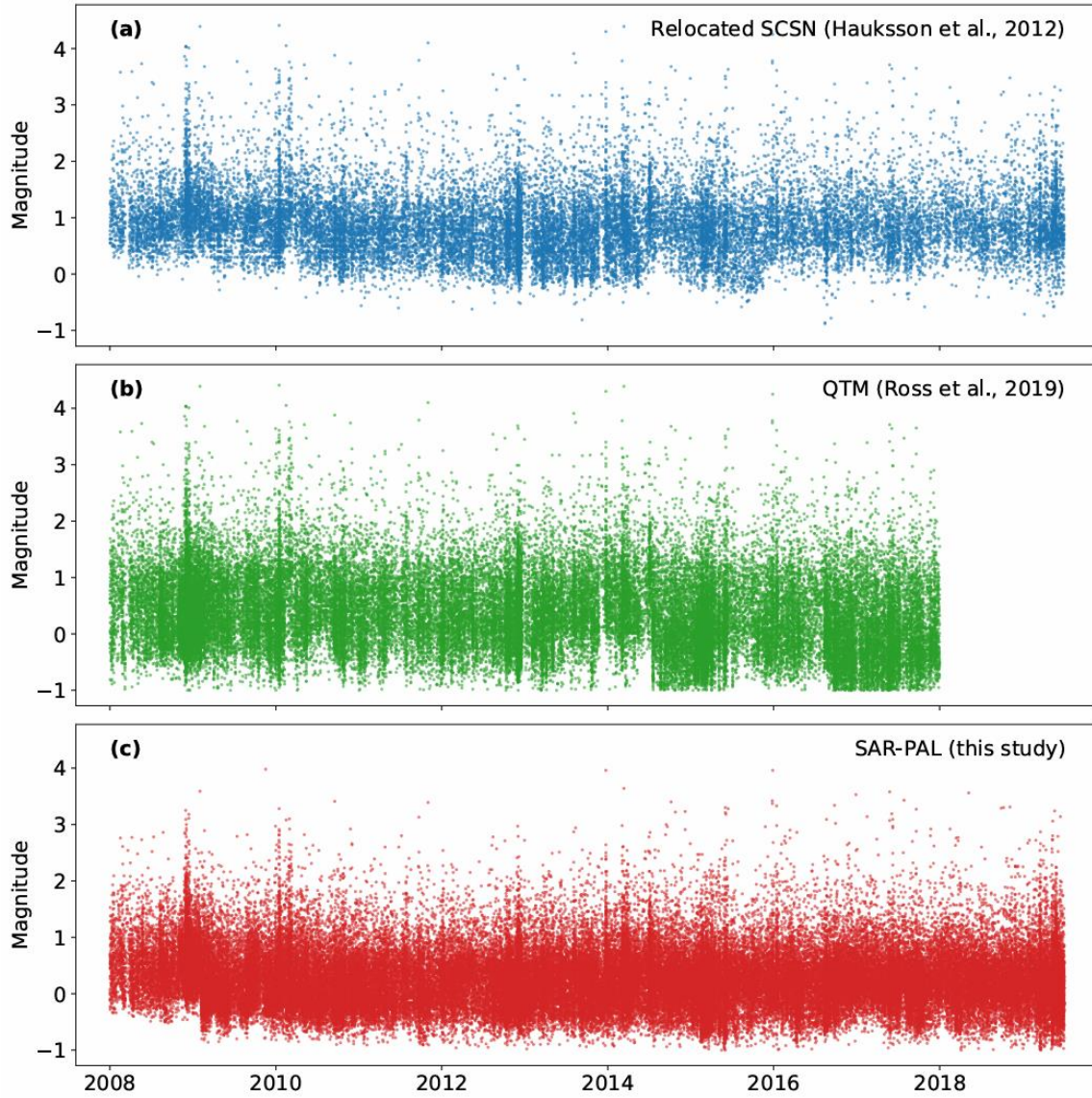


Figure S14. Magnitude-time sequence for Ridgecrest-Coso preseismic period (2008-2019/06), same as Figure 5d-f, but plot in a more complete time range. (a-c) plot the relocated SCSN catalog, well-located QTM catalog, and relocated SAR-PAL catalog, respectively.

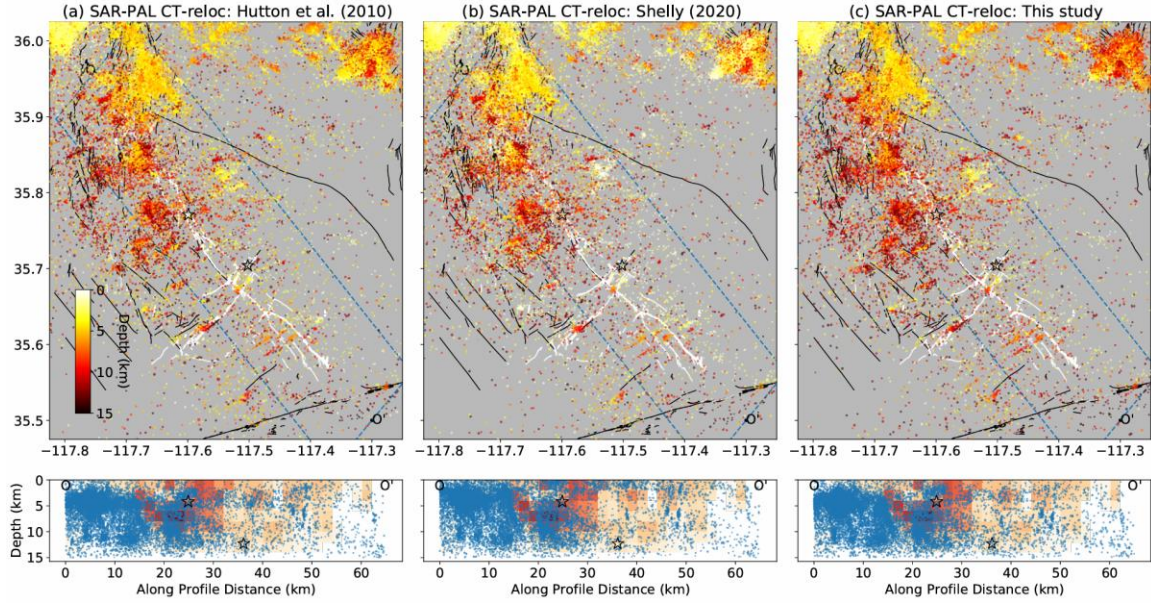


Figure S15. Comparison of SAR-PAL relocation results with different velocity models. (a-c) plot the hypoDD relocations with *dt.ct* data and the velocity models from (a) Hutton et al. (2010) for the whole Sothern California, (b) Shelly (2020) for the Ridgecrest source region, and (c) the simplified CVM-S4 model used in this study for the RC-Coso region.

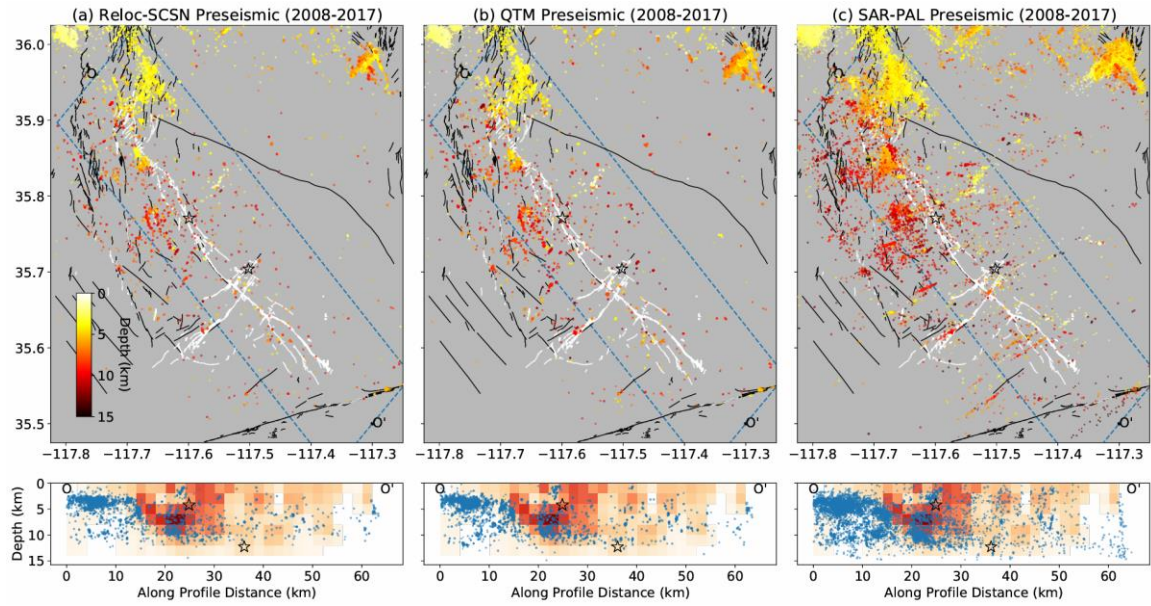


Figure S16. Distribution of Ridgecrest seismicity before the 2019 mainshock. (a-c) plot the relocated SCSN catalog, well-located QTM catalog, and relocated SAR-PAL catalog, respectively.

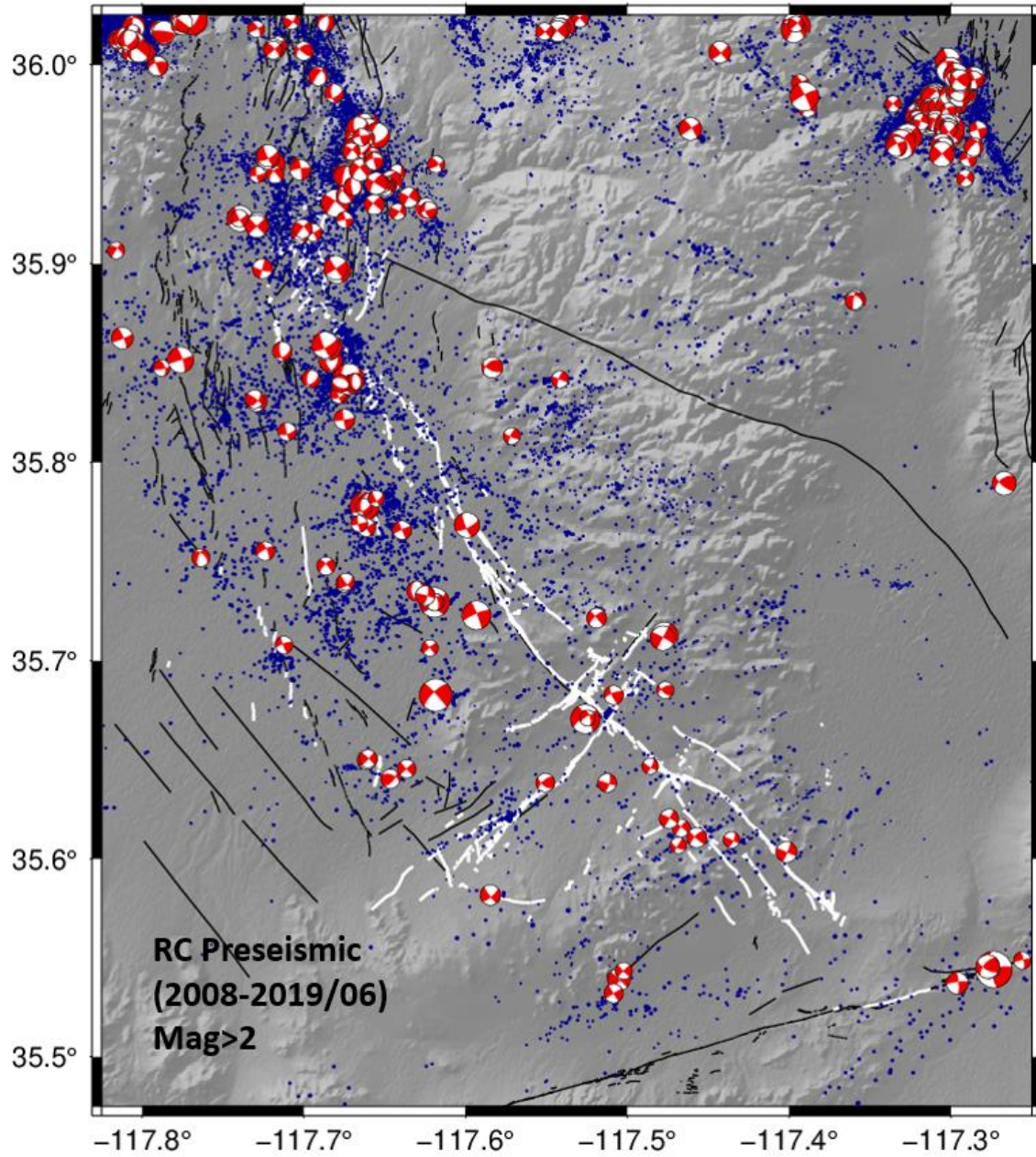


Figure S17. Focal mechanism solutions (FMS) for the Ridgecrest-Coso preseismic period (2008-2019/06). The red beach balls and dark blue dots denote the $M > 2$ FMSs from Cheng et al. (2023) and relocated SAR-PAL catalog present in this study.

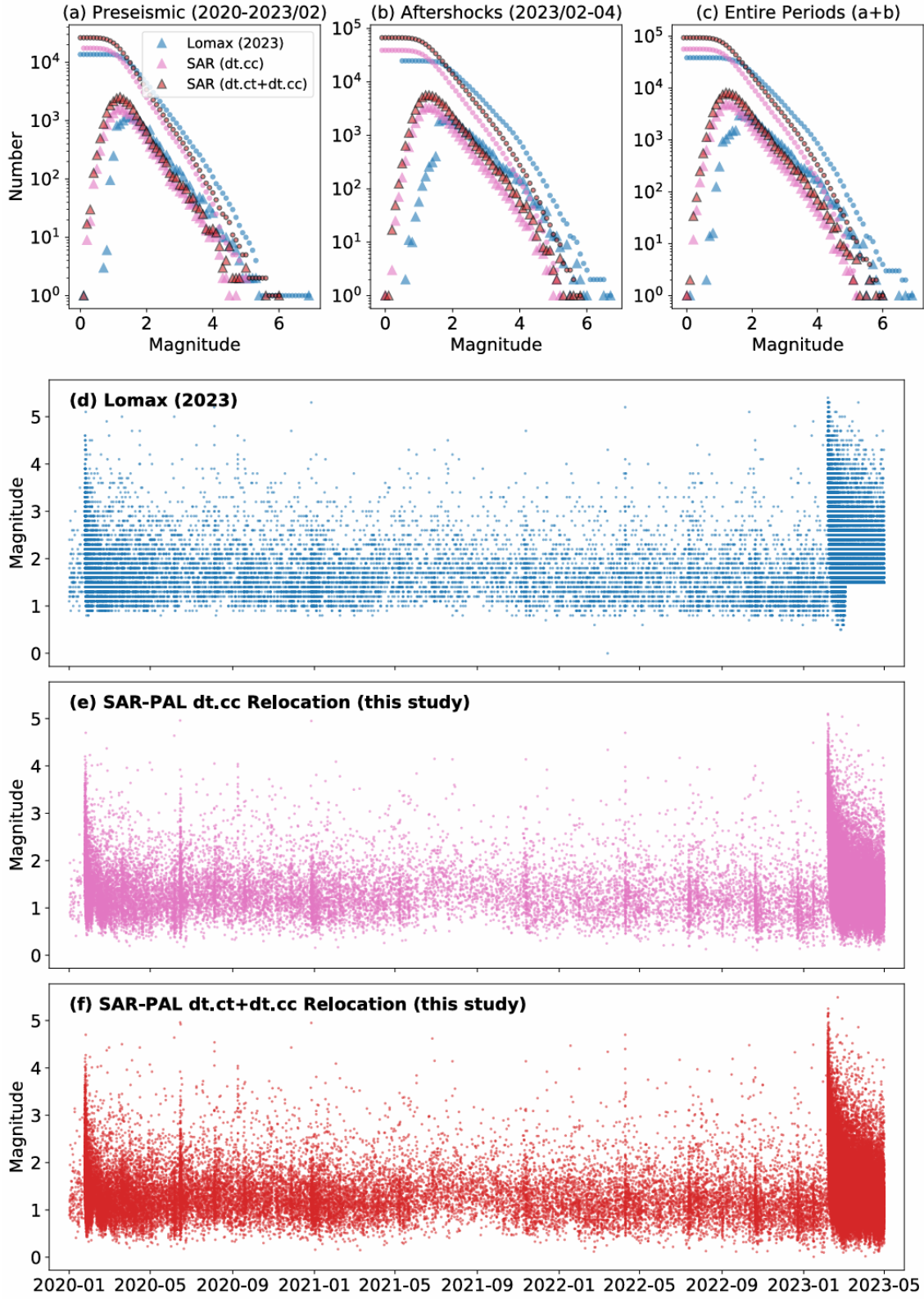


Figure S18. FMD comparison and magnitude-time sequence for EAFZ pre-postseismic period. Markers have the same meaning as Figure 5.

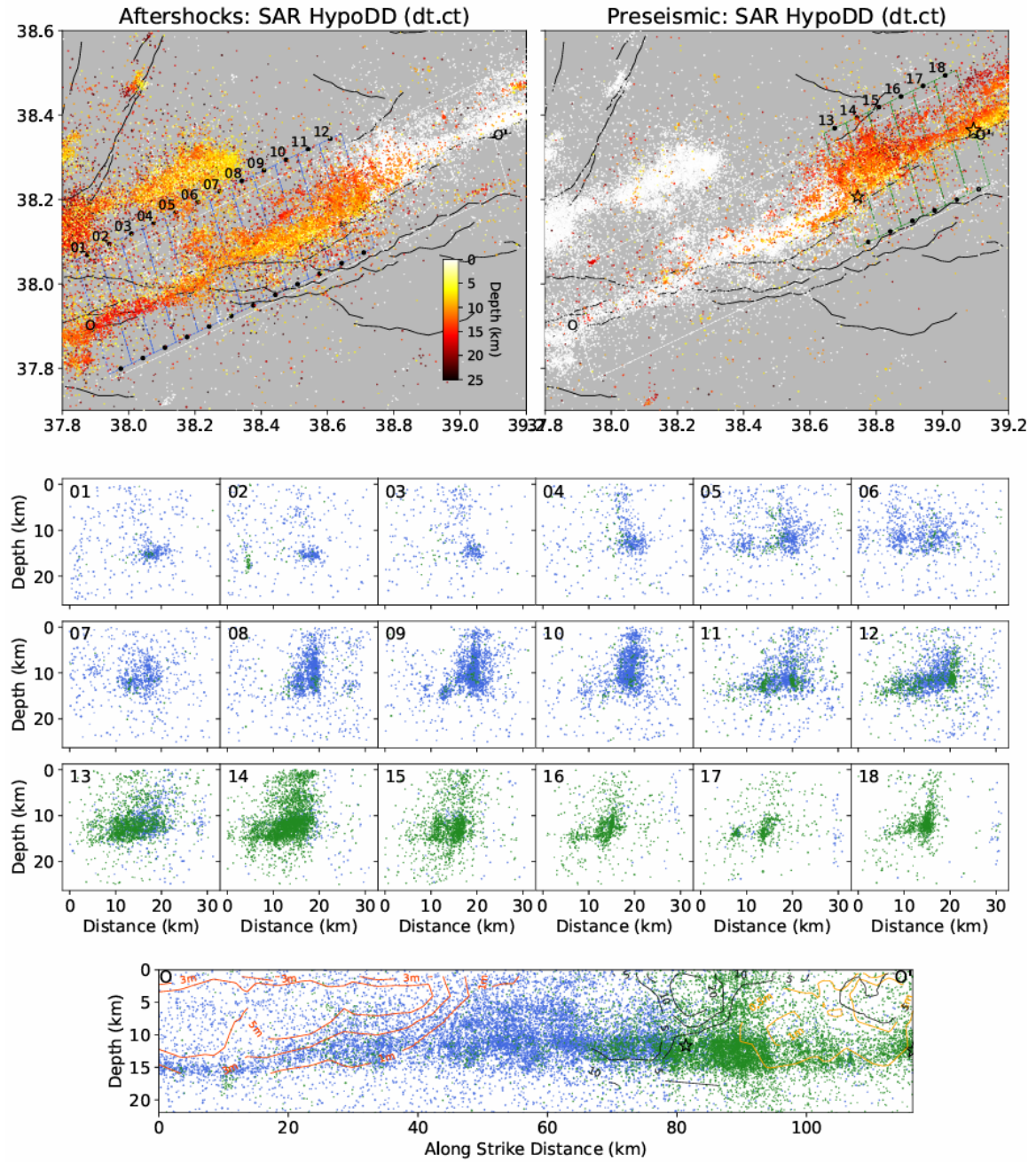


Figure S19. Seismicity distribution of EPF with SAR-PAL catalog relocated with *dt.ct*. Markers are in the same meaning as in Figure 10.

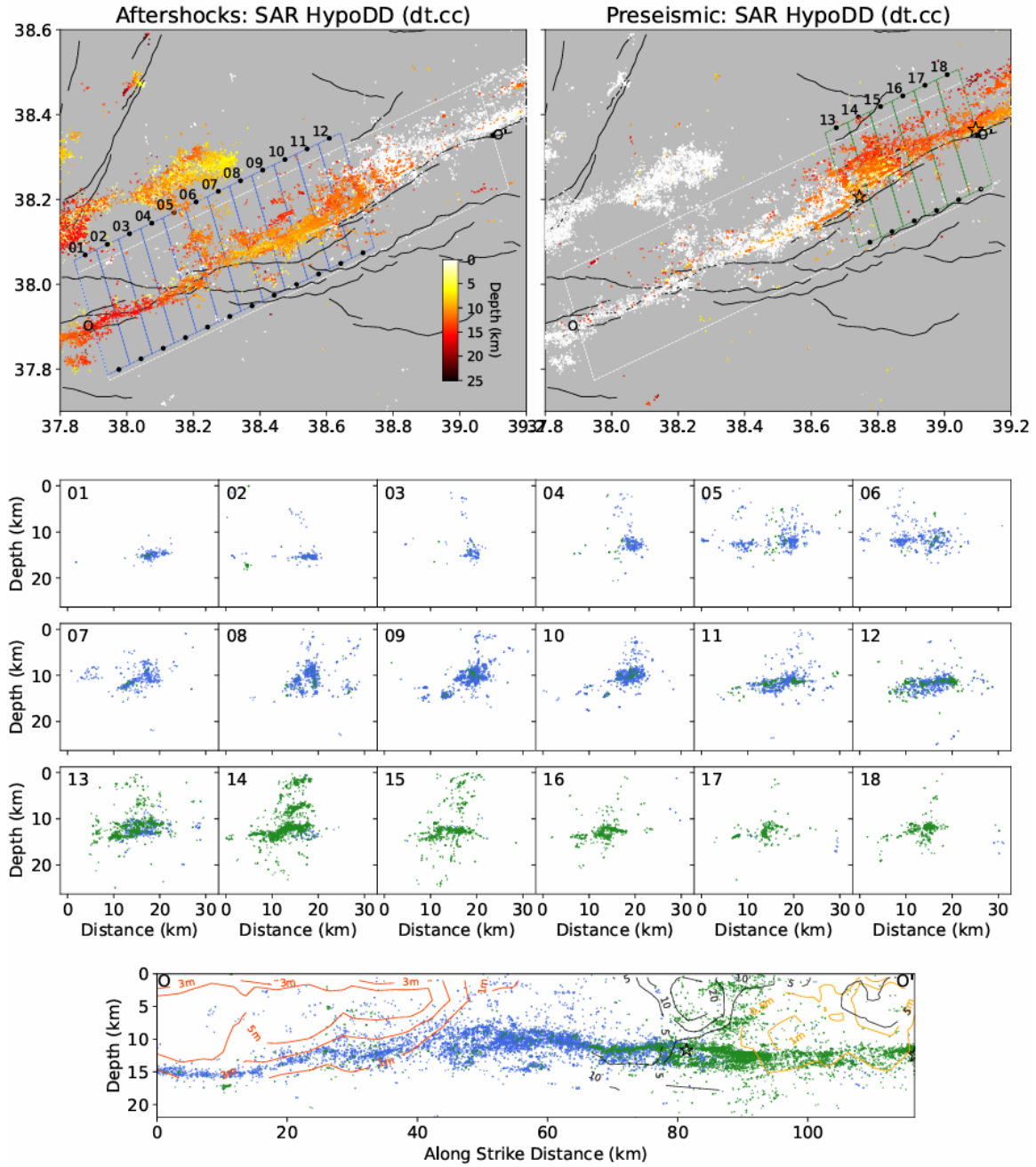


Figure S20. Seismicity distribution of EPF with SAR-PAL catalog relocated with *dt.cc*. Markers are in the same meaning as in Figure 10.

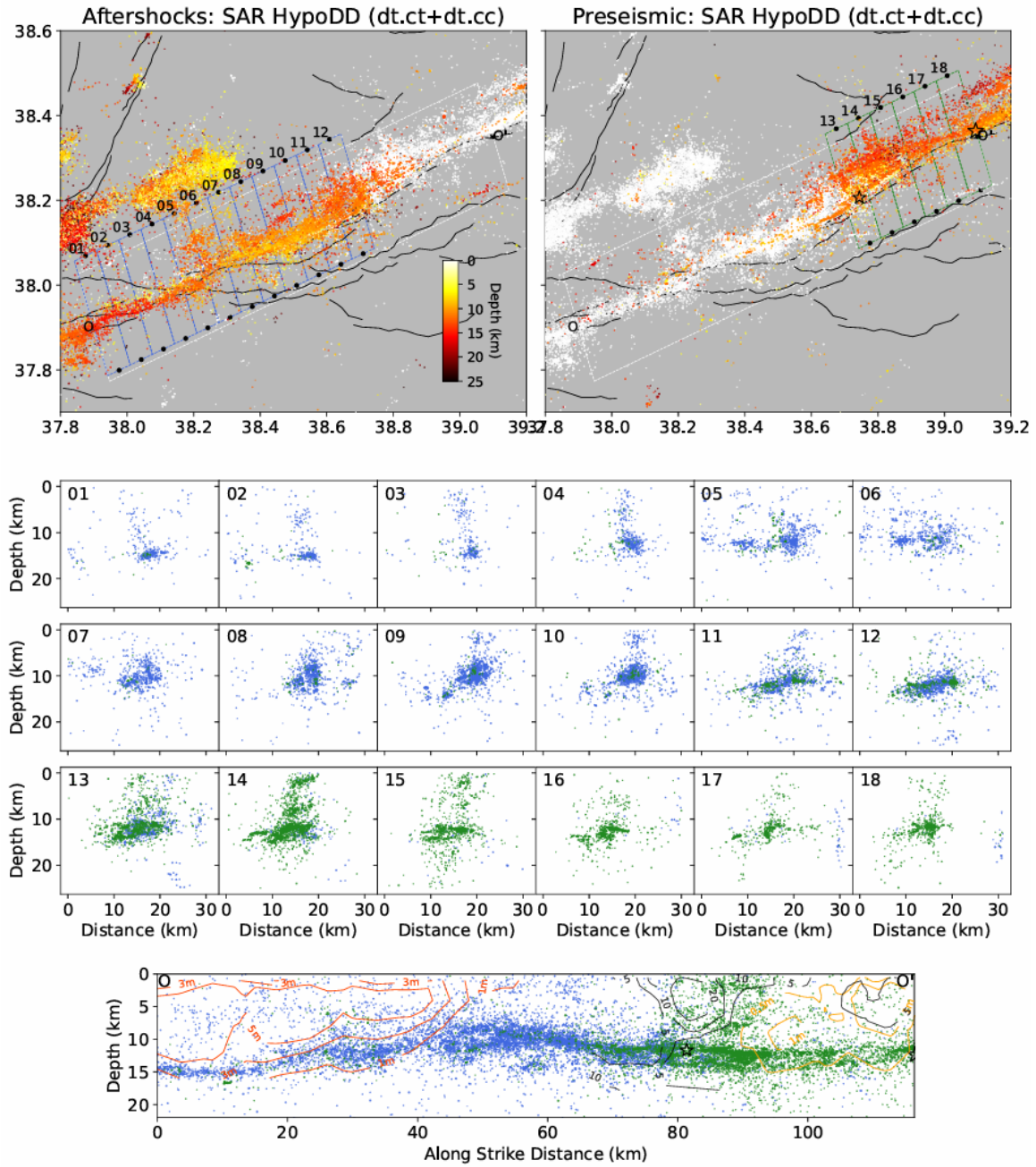


Figure S21. Seismicity distribution of EPF with SAR-PAL catalog relocated with $dt.ct+dt.cc$. Markers are in the same meaning as in Figure 10.

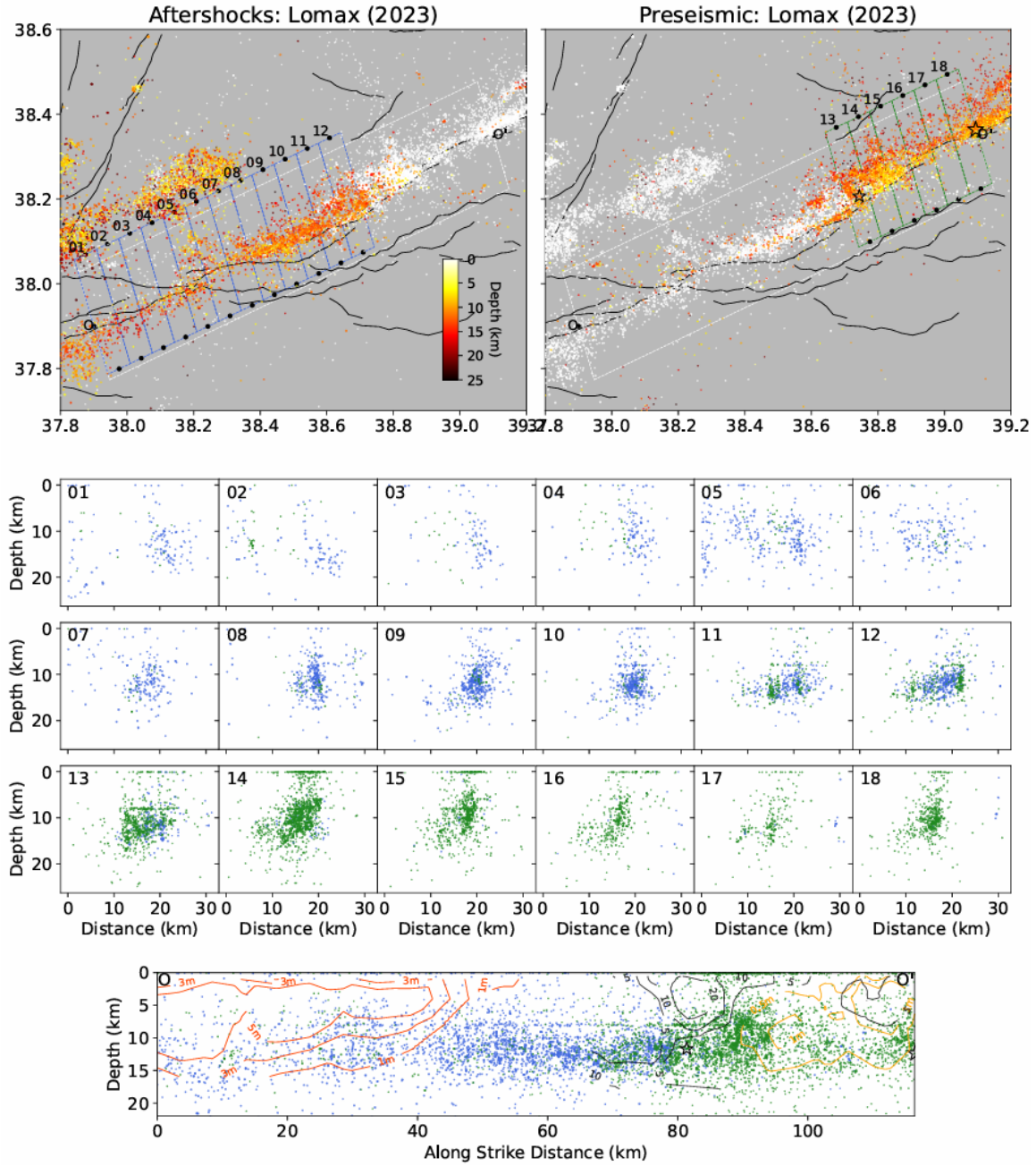


Figure S22. Seismicity distribution of EPF with relocated AFAD catalog by Lomax (2023). Markers are in the same meaning as in Figure 10.

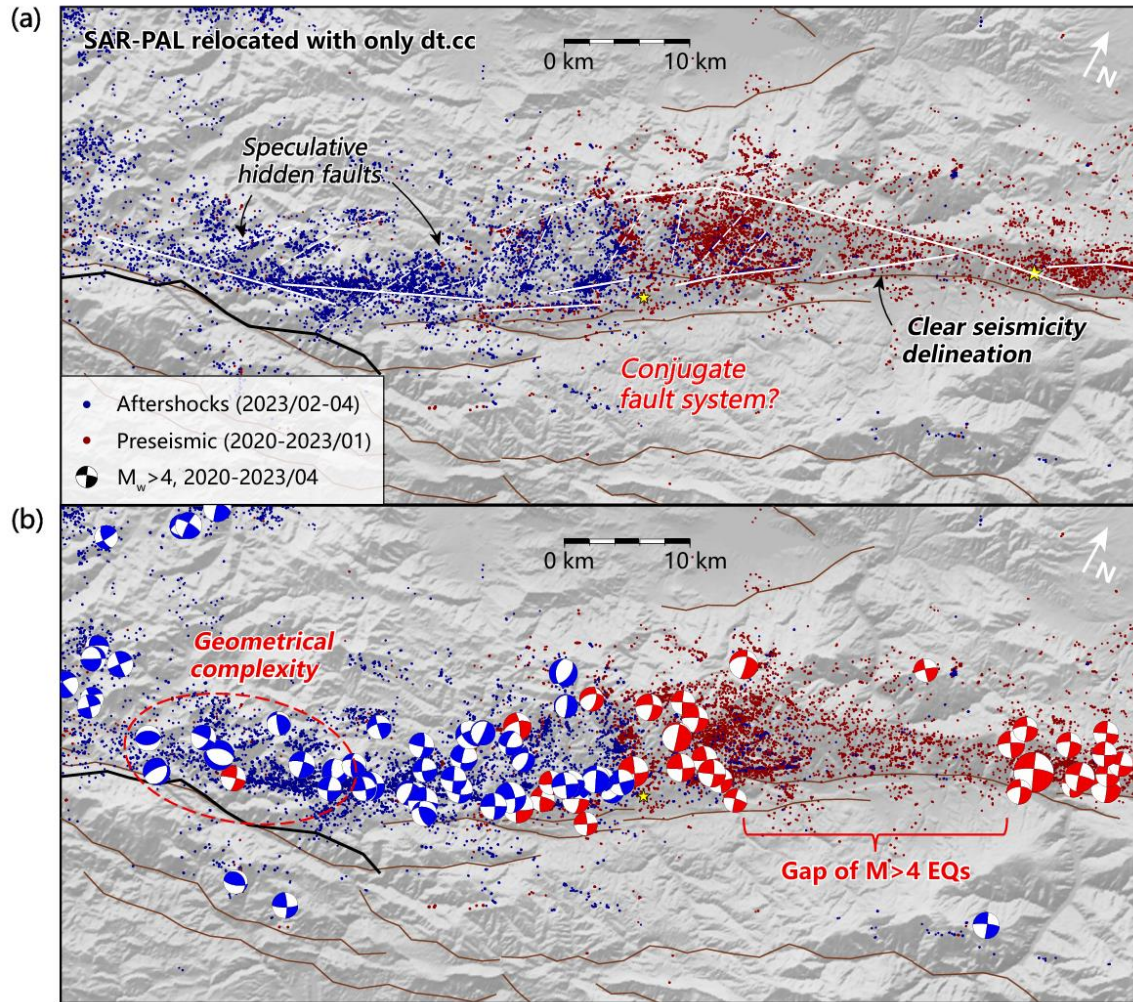


Figure S23. Detailed seismicity distribution and focal mechanisms along EPF. (a) plot the distribution of CC-relocated SAR-PAL catalog. The 2023 aftershocks and preseismic events are denoted by blue and red dots, respectively. The white solid and dashed lines show interpreted faults with clear seismicity delineation (high confidential) and those with rather scattered seismicity (speculative), respectively. (b) plot the focal mechanism solutions of $M_w > 4$ events during 2020-2023/04 from AFAD.

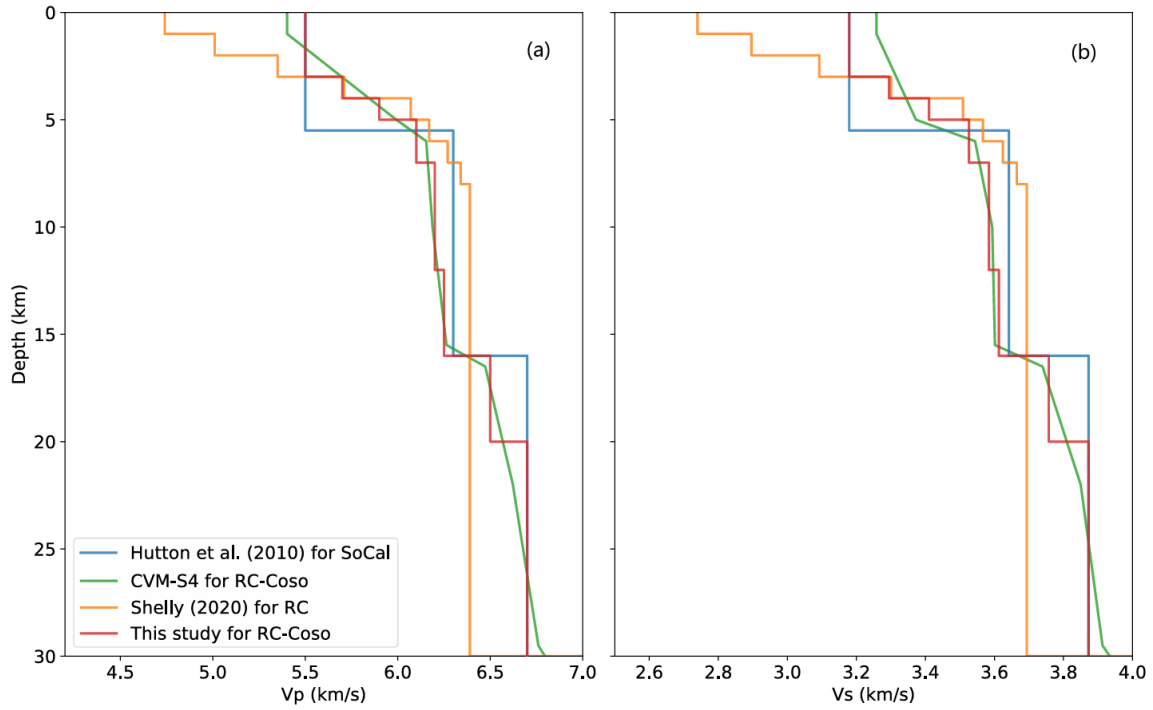


Figure S24. Comparison of velocity models for the Ridgecrest-Coso region. (a-b) plot P and S wave velocity. The blue, orange, green, and red lines denote the velocity model from Hutton et al. (2010) for the whole Sothern California, Shelly (2020) for Ridgecrest source region, Community Velocity Model - SCEC version 4 (CVM-S4) averaged in the RC-Coso region, and the velocity model used in this study, respectively.

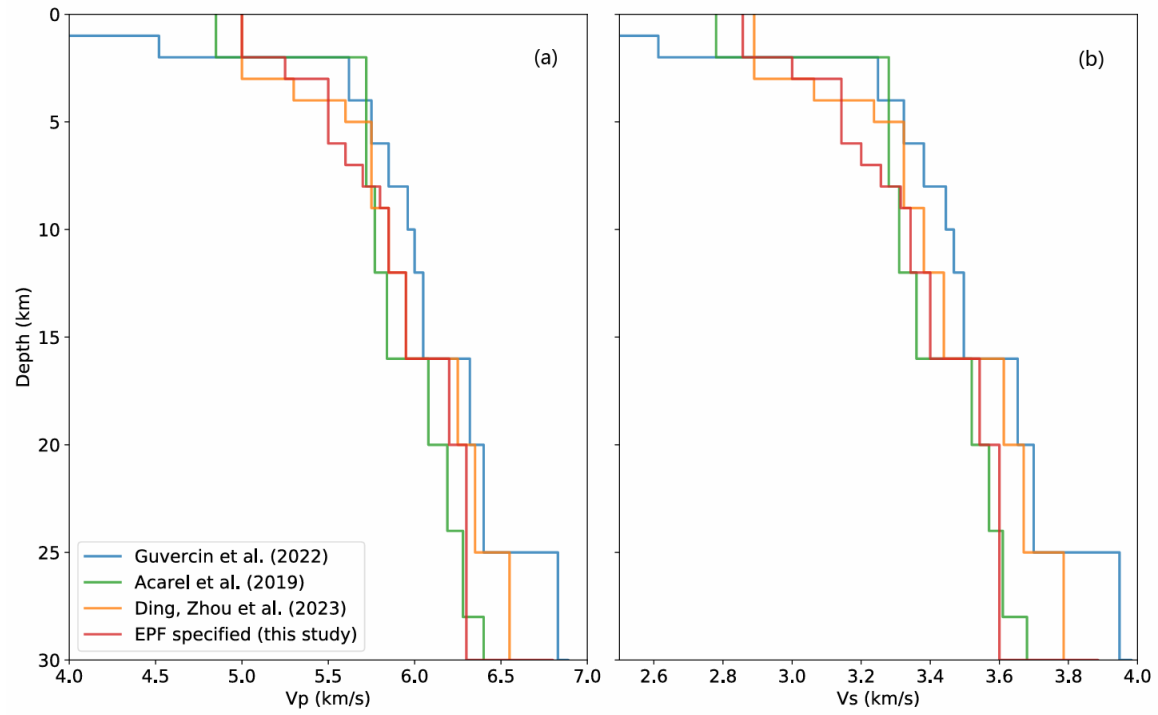


Figure S25. Comparison of velocity models for the East Anatolian Fault Zone. (a-b) plot P and S wave velocity. The blue, orange, green, and red lines denote the velocity model from Güvercin et al. (2022), Acarel et al. (2019), Ding et al. (2023), and the velocity model used in this study, respectively.

Table S1. Velocity model for Ridgecrest-Coso region used in this study.

Top of layer (km)	P-wave velocity (km/s)
0	5.50
3	5.70
4	5.90
5	6.10
7	6.20
12	6.25
16	6.50
20	6.70
32	7.80

Note: $V_p/V_s = 1.73$; depth relative to ground elevation (1500 m)

Table S2. Velocity model for EAFZ used in this study, specifically tuned for the EPF.

Top of layer (km)	P-wave velocity (km/s)
0	5.00
2	5.25
3	5.50
6	5.60
7	5.70
8	5.80
9	5.85
12	5.95
16	6.20
20	6.30
30	6.80
33	7.30
36	7.80

Note: $V_p/V_s = 1.75$; depth relative to ground elevation (1500 m)

Data Set S1. SAR-PAL seismic catalog for the Ridgecrest-Coso region preseismic period (20080101-20190703).

Data Set S2. SAR-PAL seismic catalog for the Ridgecrest-Coso region early aftershock period (20190704-20190724).

Data Set S3. SAR-PAL seismic catalog for the East Anatolian Fault Zone pre-postseismic period (2020-2023/04) relocated with *dt.ct+dt.cc*.

Data Set S4. SAR-PAL seismic catalog for the East Anatolian Fault Zone pre-postseismic period (2020-2023/04) relocated with only *dt.cc*.

References

- Acarel, D., M. D. Cambaz, F. Turhan, A. K. Mutlu, and R. Polat (2019). Seismotectonics of Malatya Fault, Eastern Turkey, *Open Geosciences*, 11(1), 1098-1111, doi:doi:10.1515/geo-2019-0085.
- Cheng, Y., E. Hauksson, and Y. Ben-Zion (2023). Refined Earthquake Focal Mechanism Catalog for Southern California Derived With Deep Learning Algorithms, *Journal of Geophysical Research: Solid Earth*, 128(2), e2022JB025975, doi:<https://doi.org/10.1029/2022JB025975>.
- Ding, H., Y. Zhou, Z. Ge, T. Taymaz, A. Ghosh, H. Xu, T. S. Irmak, and X. Song (2023). High-resolution seismicity imaging and early aftershock migration of the 2023 Kahramanmaraş (SE Türkiye) $M_{\text{W}}7.9$ & 7.8 earthquake doublet, *Earthquake Science*, 36, 1-16, doi:10.1016/j.eqs.2023.06.002.
- Güvercin, S. E., H. Karabulut, A. Ö. Konca, U. Doğan, and S. Ergintav (2022). Active seismotectonics of the East Anatolian Fault, *Geophysical Journal International*, 230(1), 50-69, doi:10.1093/gji/ggac045.
- Hauksson, E., C. Yoon, E. Yu, J. R. Andrews, M. Alvarez, R. Bhadha, and V. Thomas (2020). Caltech/USGS Southern California Seismic Network (SCSN) and Southern California Earthquake Data Center (SCEDC): Data Availability for the 2019 Ridgecrest Sequence, *Seismological Research Letters*, 91(4), 1961-1970, doi:10.1785/0220190290.
- Hutton, K., J. Woessner, and E. Hauksson (2010). Earthquake Monitoring in Southern California for Seventy-Seven Years (1932–2008), *Bulletin of the Seismological Society of America*, 100(2), 423-446, doi:10.1785/0120090130.
- Shelly, D. R. (2020). A High - Resolution Seismic Catalog for the Initial 2019 Ridgecrest Earthquake Sequence: Foreshocks, Aftershocks, and Faulting Complexity, *Seismological Research Letters*, 91(4), 1971-1978, doi:10.1785/0220190309.
- Zhou, Y., H. Yue, L. Fang, S. Zhou, L. Zhao, and A. Ghosh (2021). An Earthquake Detection and Location Architecture for Continuous Seismograms: Phase Picking, Association, Location, and Matched Filter (PALM), *Seismological Research Letters*, 93(1), 413-425, doi:10.1785/0220210111.